

Name of Student: Vyankatesh Suresh Rohokale	PRN:26070122518
Semester: III	Year : AY 2026-27
Subject: Programming with Java	
ASSIGNMENT No : 2	
TITLE : Constructors in Java	

Problem Statement

Develop a Java application demonstrating the use of default, parameterized, and copy constructors for object initialization and data management.

Learning Objectives

To understand and implement default, parameterized, and copy constructors for object initialization.

Programme

1. Create a **Student class** using default and parameterized constructors to initialize student name and roll number.
2. Develop a Mobile Phone Inventory System using different constructors to initialize mobile details and create duplicate object records.

Programme :

1 :

```
J Demo.java
1  class Student {
2
3      String name;
4      int rollNo;
5
6      // Default Constructor
7      Student() {
8          name = "Vyankatesh";
9          rollNo = 101;
10     }
11
12     // Parameterized Constructor
13     Student(String name, int rollNo) {
14         this.name = name;
15         this.rollNo = rollNo;
16     }
17
18     void display() {
19         System.out.println("Name: " + name);
20         System.out.println("Roll Number: " + rollNo);
21         System.out.println();
22     }
23
24     public static void main(String[] args) {
25
26         Student s1 = new Student();
27         Student s2 = new Student("Rahul", 102);
28
29         s1.display();
30         s2.display();
31     }
32 }
```

Output :

```
● piyushrohokale@Piyushs-MacBook-Air assignment06 % javac Demo.java && java Demo
Data written successfully.

Employee Details:
Employee ID: 518
Employee Name: Piyush
Department: Finance%
○ piyushrohokale@Piyushs-MacBook-Air assignment06 %
```

2:

```
J Demo.java
1  class Mobile {
2
3      String brand;
4      String model;
5      double price;
6
7      // Default Constructor
8      Mobile() {
9          brand = "Samsung";
10         model = "Galaxy A15";
11         price = 18000;
12     }
13
14     // Parameterized Constructor
15     Mobile(String brand, String model, double price) {
16         this.brand = brand;
17         this.model = model;
18         this.price = price;
19     }
20
21     // Copy Constructor
22     Mobile(Mobile m) {
23         this.brand = m.brand;
24         this.model = m.model;
25         this.price = m.price;
26     }
27
28     void display() {
29         System.out.println("Brand : " + brand);
30         System.out.println("Model : " + model);
31         System.out.println("Price : ₹" + price);
32         System.out.println();
33     }
34 }
```

```

34
35     public static void main(String[] args) {
36
37         // Default constructor
38         Mobile m1 = new Mobile();
39
40         // Parameterized constructor
41         Mobile m2 = new Mobile("Apple", "iPhone 15", 60000);
42
43         // Copy constructor
44         Mobile m3 = new Mobile(m2);
45
46         System.out.println("Mobile 1:");
47         m1.display();
48
49         System.out.println("Mobile 2:");
50         m2.display();
51
52         System.out.println("Duplicate Mobile Record:");
53         m3.display();
54     }
55 }

```

Output :

```

● piyushrohokale@Piyushs-MacBook-Air assignment06 % javac Demo.java && java Demo
Data written successfully.

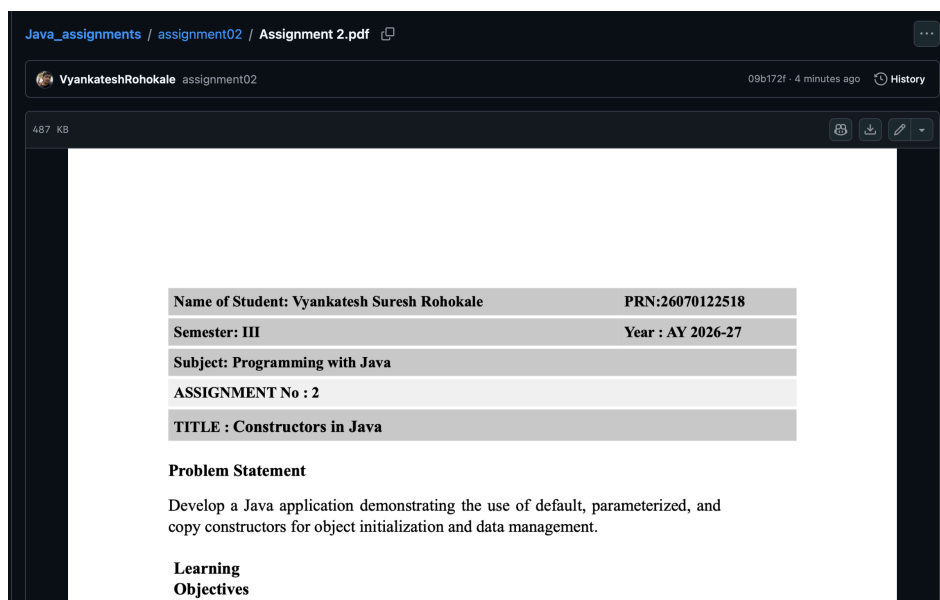
Employee Details:
Employee ID: 518
Employee Name: Piyush
Department: Finance
○ piyushrohokale@Piyushs-MacBook-Air assignment06 %

```

Conclusion

The program successfully demonstrates the use of default, parameterized, and copy constructors in Java. The default constructor initializes objects with predefined values, the parameterized constructor initializes objects with user-defined values, and the copy constructor creates a new object by copying the data from an existing object. This shows how constructors are used for object initialization and efficient data management in Java.

Github :



The screenshot shows a PDF document titled "Assignment 2.pdf" by user "VyankateshRohokale". The document contains the following information:

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